

Development of a Game System to Provide Enjoyable Playful Experiences to People in All Stages of Dementia

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ABSTRACT

In this work in progress paper, we discuss our work to develop an adaptive game system for people with dementia. The system makes use of projection-based AR and natural user interfaces to present game content in a manner which is engaging and allows for meaningful interaction. As the first step, we designed 7 different games, each based on a specific play experience. Later, we aim to investigate if each of these experiences could be enjoyed by users with varying levels of cognitive impairments.

Index Terms: Applied computing—Computer games; Social and professional topics—Seniors; Human-centered computing—Mixed / augmented reality

1 INTRODUCTION

As digital games continue to grow in popularity, their audience has become increasingly more diverse, with people ranging from young children to older adults regularly playing games as part of everyday life. However, few games in the market today could be fully enjoyed by individuals with severe levels of sensory or cognitive impairments. This is especially the case for people living with dementia, particularly those in the later stages of the condition. Such users often struggle to engage with digital games not only due to the difficulties of interacting with a traditional gaming interface (e.g. controlling the games through a mouse-and-keyboard or a joystick interface) [4, 16], but also due to the design of the games themselves and their underlining game mechanics.

The mechanics that have made games so enjoyable to conventional users, for instance, the sense of progression and accomplishment from either seeing ones characters "level up" and grow in strength or from being able to complete easier "game levels" and move on to more challenging ones, might not be as meaningful to people with dementia due to various aspects related to their cognitive decline. For example, people with dementia who experience short term memory impairments might not remember the effort they have put in to level up their character or to progress to their current game level from their previous game sessions and as such could find it difficult to experience a sense of achievement from their in-game progression as they continually play the game. In fact, a prior study had hypothesized that out of several playful experiences (e.g. challenge, competition) [11], experiences such as relaxation, reminiscence and sensation have the most potential to be effective for people with late stage dementia [2]. However, this has yet to be proven empirically as few studies have been carried out to investigate in-depth what experiences could be cognitively understood and enjoyed by people in different stages of dementia. Most studies in this domain tend to focus on the design and evaluation of games aimed at improving a specific cognitive skill or

physical ability and not on examining the general effectiveness of certain game elements [9, 13, 20]. In addition, such studies also tend to target people with mild levels of dementia and not those with high levels of cognitive impairments [12, 19]. This is unfortunate as such users tend to lack other avenues for interactive entertainment and could benefit greatly from games in general (e.g. to distract and possibly alleviate behaviors that challenge). Thus, knowledge on how games could be designed for people with late stage dementia would be especially valuable to the research community.

To address this gap in research, our study aims to investigate whether and to what degree certain playful elements (Nurture, Stimulation, Relaxation etc.) could be enjoyed by people in different stages of dementia. Building upon what was learnt from past studies [1, 16, 17], we utilize a form of projection based AR and natural user interfaces to present the game content to people with dementia in a manner which is engaging enough to sustain their attention for prolonged periods and simple enough to allow for meaningful user interactions. In particular, the system and games which we developed incorporated several suggestions from previously proposed design guidelines [6]. For example, to address issues related to declines in perceptual ability (such as vision and hearing based declines), in our games, we used big easy to see objects when possible, that are projected directly in front of users. To address issues related to motor skill declines, we designed our interaction system based on extremely simple arm movements (PwD could reach out and touch at various in-game objects in front of them using their right arm) instead of using complex arm movements (e.g. moving ones arms in a specific gesture such as in a boat rowing motion). In addition, previous studies also suggested that the environment and activities in the games should also be based on daily life activities of PwD [8] and as such, when possible, we designed the games to replicate such activities or attempted to base them on concepts PwD tend to already be familiar with (riding bicycles, fishing)

Although, the scope of our current study is to evaluate what playful experiences are effective with different types of cognitive impairments, our eventual goal is to develop an adaptive game system that is able to automatically personalize the game mechanics to each user based on their cognitive capacity. The expected outcome of our game system is that they not only provide an enjoyable enough past-time for people in late stage of dementia to help improve their emotional well-being and reduce behaviours that challenge such as aggression and agitation (similar to other interactive systems [15]), but also provide such users, who tend to live sedentary lives [2], with small opportunities for physical movement and light exercise. Overall, based on the criteria proposed in a previous study outlining the use of AR in serious games, the game system which we designed utilizes AR in the role of "helping convey serious content for each individual case" [14].

2 DEVELOPMENT OF THE GAME SYSTEM

Our prior studies have shown that people in the later stages of dementia tend to lack the cognitive skills (e.g. reasoning and deduction ability) necessary to understand the complex navigation structures found in conventional game systems as well as the fine-motor skills needed to engage with controller devices used in conventional gaming inter-

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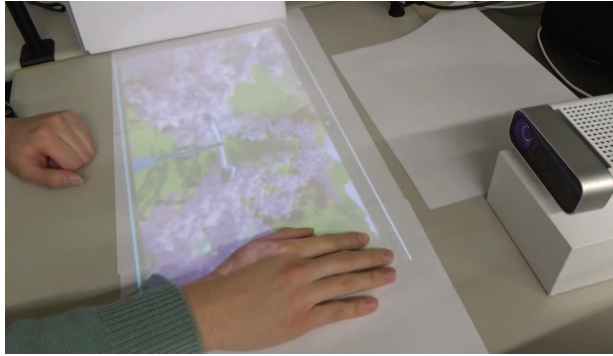


Figure 1: The setup and devices used by our dementia game system.

faces (e.g. due to the weakened control of limbs, etc.) [16, 17]. In addition, such users tend to lose their attention quickly when multimedia content is shown on a distant surface such as a TV screen or computer monitor [16]. To address such issues, we have developed a system in which our games are projected directly in front or around the user, either on the table, nearby wall or on the floor around the user using a small form digital projector. This was done to immediately grab the attention of the user and help sustain their attention during game play. To reduce the complexity of interaction for such users, we turn to natural user interfaces and use gesture recognition sensors (i.e. the Azure Kinect) to allow users to engage directly with the projected game content in an easy to understand manner: by swiping at them or touching them directly. This was achieved by using the highly accurate skeleton tracking feature from the Kinect Sensor to detect arm movements from the users [18] and then mapping them onto the virtual space to detect whether or not they collided with the virtual game objects. To further improve the reliability of the gesture input, we also made use of the depth data available from the Kinect sensor when detecting motion. The use of the depth camera along with a body tracking model that was trained using data from people in a windowless dark room (to exclude influences of external light) [3], allows our system to function well in the low light conditions that is needed when a projector is used. Compared to existing touchscreen-based methods or smart gadgets [5, 7], the system setup which we used offer more flexibility and allows us to more easily replicate various environments and immersive experiences (bicycle riding etc) at the expense of the effort needed to setup the system. In addition, using a projection device instead of a tablet screen allows us to project a larger interaction space which helps compensate for the perceptual and motor skills declines commonly encountered by PwD. Figure 1 shows the setup and devices used by our game system.

3 DESIGNING THE SEVEN PLAY EXPERIENCES

According to existing literature, twenty-two types of play experiences were reported [10]. However, based on Neuropathological and clinical studies, a total of eleven play experiences were recommended by the researchers for people with moderate to severe Alzheimer's Disease [2]. These include challenge, eroticism, expression, fellowship, humor, nurture, relaxation, reminiscence, sensation, simulation and sympathy. Out of the eleven play experiences however, four of them (eroticism, fellowship, sympathy, humour) cannot be clearly defined due to individual differences and thus, these four play experiences were difficult to gamify and distinguishing between them and the other seven is demanding. Therefore, in this study, we decided to design playful games that represent the remaining seven experiences and investigate which of them could be enjoyed by people in different stages of dementia and to what degree. We

should note that we chose to focus more on a broad range of playful experiences such as these rather than focus on more conventional strategies such as an adaptable difficulty level as we were not certain whether people with severe levels of cognitive impairments would be able to experience and understand feelings such as flow [2]. To ensure that the games are appropriate for people with Dementia, we attempted to 1) make the game objects as large as possible for easy interaction 2) create an intuitive playful experience by using objects, situations and actions that participants are already familiar with in daily life and 3) minimize the physical exertion of playing the games to reduce fatigue. Figure 2 shows screenshots of the seven different games. Each of them are described in detail below.

- The **relaxation play experience** was designed to provide participants with a peaceful experience. For this experience, we created a virtual pond filled with small fish. Participants would be able to watch fish swim in the pond in a relaxing manner and listen to soothing music. It should be noted that as we did not want to stress users, they were not provided with a set task or goal in this experience.
- The **reminiscence play experience** was designed to allow people with dementia to experience memories when they were younger. For this experience, we projected an old-fashioned radio that played different music from the early into the century when users "touched" the radio knob. Players would be successful if they touch the right knob to play their preferred music.
- For the **fantasy play experience**, we think of fantasy games as having users experience worlds or realities that isn't commonly seen in real life. Therefore, we decided on the theme of space travel and designed a game where players would attempt to steer a ship around a field of asteroids by touching onto the ship. To remove the element of challenge from the game, the ship would propel forward along a set path whenever players touch it.
- For the **challenge play experience**, we designed a fishing game. Players would attempt to collect the fish by touching on them using their hands as they swim around rapidly. Fishes of different sizes and types were included to provide different levels of difficulty. The results of the game will be judged as success or failure according to the participant's performance, with participants expected to feel a sense of accomplishment when they succeed.
- The **nurture play experience** was designed around the concept of nurturing a pet or an animal. Players are expected to feel a sense of accomplishment as they gradually see their pet grow. As such, for this experience, we designed an animal-feeding game in which the animal pet model grew bigger as participants fed them. Participants would feed the animal by touching the projected animal.
- When designing the **simulation play experience**, we wanted to make users experience a real event as part of the game. We chose to simulate the experience of cycling, which almost everyone has experienced in the country we plan to conduct our experiment. In the game, players are provided with a first person view of them riding the bicycle and could rapidly touch or hold the bicycle handle to move forward along a set path which is aligned with beautiful flowers and trees.
- For the **expression play experience** we designed a doodling and drawing game which allows users to express themselves by drawing their favorite patterns or objects which they like or dislike. The sensors were programmed to detect the motion

trajectory of the right hand of the user, thus allowing players to draw directly onto the projected surface by swiping their hands. In the game, players would be asked to draw certain objects based on their own imagination.



Figure 2: Screenshots of the seven games (Top row left: relaxation, Top row right: reminiscence, Second row left: fantasy, Second row right: challenge, Third row left: nurture, Third row right: simulation, Bottom row: expression)

To affirm whether the games we developed brought about the playful experiences they were designed to represent, we asked 5 participants to play each of the games and rate the degree in which they experienced the corresponding playful experiences on a 10-point Likert scale (min=0, max=9). It should be noted that as the main aim was to verify if the games we created met the definition of the playful experiences as defined by generic users, we asked general users without dementia to evaluate the games (Females aged 25, 38, 60 and Males aged 25 and 64). Overall, the results showed that most of the games were able to evoke the experiences they were designed to represent (relaxation mean= 5.8, sd = 2.04, reminiscence mean= 6.8, sd=1.47, fantasy mean= 8.2, sd=1.16, challenge mean=8.2, sd=1.16, nurture mean=8.2, sd=0.97, simulation mean=7.8, sd=1.93), with the only exception being expression (as participants had some difficulty using the system (mean=3.6, sd=1.62)).

4 CONCLUSION AND FUTURE WORKS

This paper describes our ongoing work to design and develop a game system which could be enjoyed by people with dementia, particularly those in the later stages of the condition. Seven different games, each representing a specific playful experience which was thought to appeal to people with moderate to high levels of cognitive impairments, were developed and would be tested with people with dementia. As part of our future work, an experiment study would be carried out where people with high levels of cognitive impairment (based on cognitive assessment scores such as the MMSE) and different types of dementia (Alzheimer's disease, Lewy Body Dementia etc.) would be asked to play each of the games. To evaluate how

enjoyable and engaging each game mechanic is, we would carry out a systematic observation of user interactions during gameplay and use observational tools such as Dementia care mapping or the Observed Emotion Rating Scale to measure changes in emotions and behaviour before and after gameplay in combination with Biometric signals such as Heart Rate, galvanic skin response and EEG.

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